

Function Operations and Composition of Functions

Algebra Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Evaluate functions at specific input values and interpret results as coordinates
- Perform operations (addition, subtraction, multiplication, division) on functions
- Find and simplify composite functions using function notation

Problems

1. Given the functions below, evaluate f of 3.

$$f(x) = 2x + 3$$

2. Using the functions below, write f of 9 equals 21 as an ordered pair.

$$f(9) = 21$$

3. Using the functions below, evaluate h of 2 minus g of 3.

$$h(x) = -3x, \quad g(x) = x^2$$

4. Using the functions below, find f of x plus g of x and simplify.

$$f(x) = 2x + 3, \quad g(x) = x^2$$

5. Using the functions below, find f of x times h of x and simplify.

$$f(x) = 2x + 3, \quad h(x) = -3x$$



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Labanan ang agwat.

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$$f(x) = 2x + 3, \quad g(x) = x^2$$

7. Find the composite function f of g of x and simplify.

$$f(x) = 2x + 3, \quad g(x) = x^2$$

8. Find the composite function g of h of x and simplify.

$$g(x) = x^2, \quad h(x) = -3x$$

9. Find the composite function f of h of x evaluated at x equals 1.

$$f(x) = 2x + 3, \quad h(x) = -3x$$

10. Using all three functions below, find f of g of h of x and simplify completely.

$$f(x) = 2x + 3, \quad g(x) = x^2, \quad h(x) = -3x$$



Function Operations and Composition of Functions — Answer Key

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Answer Key

1. Answer: $f(3) = 9$

- Replace x with 3: $f(3) = 2(3) + 3$
 - Simplify: $6 + 3 = 9$
-

2. Answer: (9, 21)

- The input value is the x -coordinate: $x = 9$
 - The output value is the y -coordinate: $y = 21$, so the ordered pair is (9, 21)
-

3. Answer: -15

- Find $h(2)$: $-3(2) = -6$
 - Find $g(3)$: $3^2 = 9$
 - Subtract: $-6 - 9 = -15$
-

4. Answer: $x^2 + 2x + 3$

- Write the sum: $(2x + 3) + x^2$
 - Arrange in standard form: $x^2 + 2x + 3$
-

5. Answer: $-6x^2 - 9x$

- Multiply: $(2x + 3)(-3x)$
 - Apply the distributive property: $-3x \cdot 2x + (-3x) \cdot 3 = -6x^2 - 9x$
-

6. Answer: $1/5$

- Set up the division: $g(1)/f(1) = 1^2/(2(1)+3)$
 - Simplify: $1/5$
-

7. Answer: $2x^2 + 3$

- Start with $f(x) = 2x + 3$ and replace x with $g(x) = x^2$
 - $f(g(x)) = 2(x^2) + 3 = 2x^2 + 3$
-

8. Answer: $9x^2$

- Start with $g(x) = x^2$ and replace x with $h(x) = -3x$
 - $g(h(x)) = (-3x)^2 = 9x^2$
-

9. Answer: -3

- Replace x in $f(x)$ with $h(x)$: $f(h(x)) = 2(-3x) + 3 = -6x + 3$
 - Evaluate at $x = 1$: $-6(1) + 3 = -6 + 3 = -3$
-



- First find $g(h(x))$: replace x in $g(x) = x^2$ with $-3x$ to get $(-3x)^2 = 9x^2$
 - Then find $f(g(h(x)))$: replace x in $f(x) = 2x + 3$ with $9x^2$ to get $2(9x^2) + 3 = 18x^2 + 3$
-

